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2. Why linear algebra?

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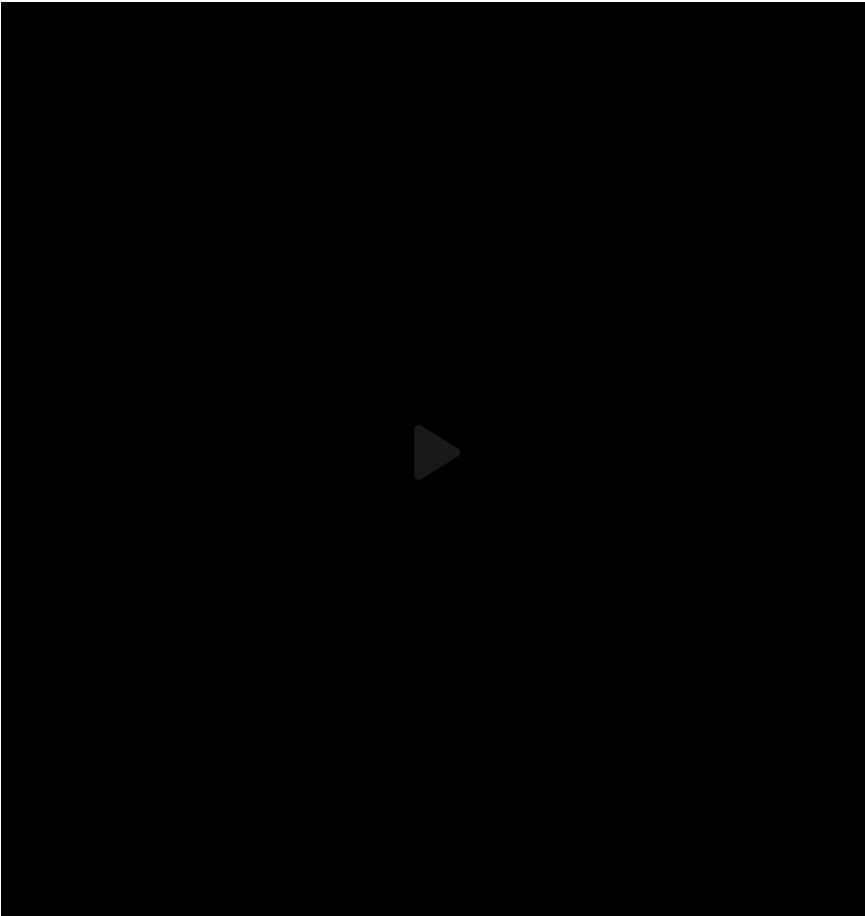
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A 3x3 example



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Why linear algebra?

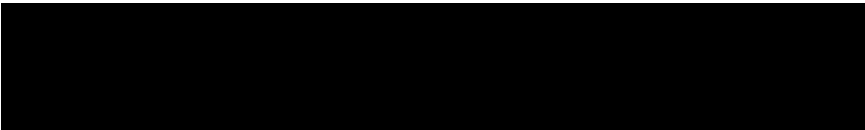
Setting the context

Systems of differential equations arise naturally from many physical scenarios. Additionally, any higher order ODE (or system of such) can be written as a system of first-order ODEs. (This is called finding the companion matrix.) This is useful because MATLAB and other computer algebra systems have routines for solving systems of first order ODEs!

Here is where linear algebra enters the stage. Let's review how linear algebra came into play in solving 2x2 systems, so we can outline the goals of our understanding of linear algebra for the context of solving differential equations.

Note on video example: We'll revisit the solution to the example in the video above in a later lecture once we've developed the machinery.

Review how to solve 2x2 systems of DEs



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...equations ...
I'll write them first out as a system.
x1 prime is-- let's take a equal 1 because I'm going to want to solve them numerically, and I want you to be able to concentrate on what's important, what's new now, and I don't want to have an extra a floating around everywhere just contributing nothing but a mild confusion to the proceedings.
So x1 prime, I'm going to take a equal 1 and simply write it minus 2 x1 plus x2 plus x3.
And so now what would the equation for x2 prime be? Well, here x2 plays the role that x1 played before. And the only way to tell you



times λ equals 0,
gives the solution is the
eigenvector α 1.

And then, the final solution
that you make out
of the two of them is α 1
looks like α 1 times
 $e^{\lambda t}$.

Of course, you do that for
each eigenvalue.

You get the associated
eigenvector,
and then the general
solution is made up
out of a linear combination
of these guys, individual
guys,

**with constant
coefficients.**



Video

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Review of solving 2x2 systems of DEs

Given a system of differential equations, our first step is to write this system as a matrix system

$$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{b}$$

For now, we assume $\mathbf{b} = \mathbf{0}$, as we will not handle solving the inhomogeneous case until we've developed tools from linear algebra to help us in this endeavor.

The process for solving the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ which has distinct eigenvalues is as follows:

Step 1: Find the eigenvalues of A . (Find λ_1, λ_2 that solves $\det(A - \lambda I) = 0$.)

Step 2: Find the eigenvectors of $\mathbf{A}$ corresponding to each eigenvalue. (Find $\mathbf{v}_1, \mathbf{v}_2$ that solve $(\mathbf{A} - \lambda_i \mathbf{I}) \mathbf{v} = \mathbf{0}$ for each $\lambda_i, i = 1, 2$.)

Step 3: Solutions take the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$$

for arbitrary constants $\mathbf{c}_1$ and $\mathbf{c}_2$ which can be determined from initial conditions.

The process of solving homogenous higher order systems of differential equations consists of the same steps if there are distinct eigenvalues. The key issues are that we must now expand our linear algebra toolbox to handle solutions to linear systems, linear independence, determinants, eigenvalues and eigenvectors, and more in 3 or more dimensions. We will also be able to use these tools to solve systems with repeated eigenvalues. That is the goal of the next several lectures.

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